REVIEW OF LITERATURE

3. REVIEW OF LITERATURE

Literature survey was carried out on the proposed topic with the facility of internet, helinet and by referring scientific journals. The survey reveals that, no work has been reported on the proposed topic and some related research works are quoted below.

1.Vinita Singh *et al.*, (2023) evaluated the efficacy and safety of diclofenac sodium patches specifically in treating dysmenorrhea, drawing insights from available literature. Clinical trials and studies have demonstrated the effectiveness of diclofenac sodium patches in providing sustained pain relief and improving menstrual symptoms associated with dysmenorrhea. By delivering the drug transdermally, these patches offer targeted relief directly to the site of pain, bypassing gastrointestinal metabolism and potentially minimizing systemic adverse effects. Additionally, the convenience and ease of application make diclofenac sodium patches a favourable option for women seeking non-oral alternatives for managing menstrual pain. Despite the generally favourable safety profile observed in studies, careful monitoring is warranted to mitigate potential adverse effects, particularly skin irritation or allergic reactions. Overall, diclofenac sodium patches emerge as a promising and well-tolerated therapeutic.[51]

2.Neha Rathore *et al.*, (2023) reported on the pathogenesis of rheumatoid arthritis focused on auto antibodies and immune complexes. Rheumatoid arthritis is the most common inflammatory arthritis and is a major cause of disability. T-cell-mediated antigenspecific responses, T-cell-independent cytokine networks, and aggressive tumourlike behaviour of rheumatoid synovial have also been implicated. Based on the pathogenic mechanisms, specific therapeutic interventions can be designed to suppress synovial inflammation and joint destruction in rheumatoid arthritis. “Diclofenac sodium” is a non-steroidal anti- inflammatory drug (NSAIDs) advocated for use in painful and inflammatory rheumatic and certain non-rheumatic conditions. It is available in a number of administration forms which can be given orally, rectally, topically or intramuscularly. The main objective of this study was

to prepare the optimized formulation of Diclofenac sodium transdermal patch for the treatment of Rheumatoid arthritis. This study is further aimed to analyse, concentration of drug reaching in the body and to study its effect.[52]

3.Priyanka Kriplani *et al.,* (2022) investigated on Transdermal drug delivery has made an important contribution to medical practice. An advantage of a transdermal drug delivery route over other types of medication delivery is that the patch provides a controlled release of the medication into the patient, usually through either a porous membrane covering a reservoir of medication or through body heat melting thin layers of medication embedded in the adhesive. The present investigation was aimed to formulate transdermal films of non-steroidal anti-inflammatory drug, Diclofenac sodium using mercury substrate method and evaluated for physicochemical parameters like thickness, weight variation, moisture uptake, moisture content, folding endurance and drug content values. Three transdermal patches were prepared using different concentrations of ethyl cellulose. It was concluded that as the concentration of polymer increases the thickness of patch, weight uniformity and folding endurance increases. Percentage moisture content and percentage moisture uptake decreases with increase in polymer concentration.[53]

4.Dandigi M *et al.,* (2021) developed and characterized Diclofenac Diethylamine (DDEA) transdermal patch using Silicone and acrylic adhesives combination. Modified solvent evaporation method was employed for casting of film over Fluoropolymer coated polyester release liner. Initial studies included solubilization of drug in the polymers using solubilizers. The formulations with combination of adhesives were attempted to combine the desirable features of both the adhesives. The effect of the permeation enhancers on the drug permeation was studied using pig ear skin. All the optimized patches were subjected to adhesion, dissolution and stability studies. A 7-day skin irritancy test on albino rabbits and an in vivo anti-inflammatory study on wistar rats by carrageenan induced paw edema method were also performed. The results indicated the high percent drug permeation (% CDP-23.582) and low solubility nature (1%) of Silicone adhesive and high solubility

(20%) and low% CDP (10.72%) of acrylic adhesive. The combination of adhesives showed desirable characteristics for DDEA permeation with adequate % CDP and sufficient solubility. Release profiles were found to be dependent on proportion of polymer and type of permeation enhancer. The anti-inflammatory study revealed the sustaining effect and high percentage inhibition of edema of C4/OLA (99.68%). The acute skin irritancy studies advocated the non-irritant nature of the adhesives used. It was concluded that an ideal of combination of adhesives would serve as the best choice, for fabrication of DDEA patches, for sustained effect of DDEA with better enhancement in permeation characteristics and robustness.[54]

5.Kashyap Ankita *et al.,* (2020) formulated and evaluated topical gel of Ibuprofen with Carbopol 940 as the polymer. Gels were prepared by dispersing the polymers in a mixture of water and glycerol with methyl paraben as the preservative and the varying amount of ibuprofen, being kept under magnetic stirring until the homogeneous dispersion was formed. The dispersion was then neutralized and made viscous by the addition of triethanolamine. The Carbopol gel of Ibuprofen were found to be homogenous with good drug loading. The pH of all the gel formulations was found within the neutral pH range which is compatible with skin. And the viscosity of the formulations was found to be feasible for topical drug delivery. The drug content of the three formulations was found in the range of 87.56% to 90.45% which shows efficient drug loading. Results of invitro drug release study showed that F5 formulation has better diffusion of drug through egg membrane and hence further permeation studies were carried out through rat epidermis. The compatibility study showed that the major peaks in FTIR spectra of the pure drug were found to be intact in their physical mixture. Hence there is no interaction between drug and Carbopol in their physical mixture. Carbopol can be effectively used as the polymer for topical gel preparation. And F5 formulation containing 0.5 % w/w Carbopol 940 may be effectively used as topical transdermal delivery for Ibuprofen.[55]

6. G Parthasarath *et al.,* (2011) developed matrix type of transdermal patches of Naproxen with ethylcellulose and Hydroxy propyl methyl cellulose polymer in various concentrations and dibutylphthalate as plasticizer. These transdermal drug

patches were characterized for their thickness, tensile strength, content uniformity, *in-vitro* release. The release profiles were found to be varied with various concentrations of Ethyl cellulose Polymer. The sample of patches prepared with 2:8 and 8:2 ratios of Ethyl cellulose and Hydroxy Propyl Methyl Cellulose showed highest and lowest *in-vitro* release of Naproxen respectively[56].

7. Lavanya Gongulamudi *et al*., (2011) developed UV Spectroscopic method for determination of tramadol hydrochloride in bulk and formulation. The present study describes a simple, accurate, precise and cost-effective UV- Spectrophotometric method for the estimation of Tramadol Hydrochloride, an Analgesic. The drug was first dissolved in N- N-dimethylformamide and final volume was made up with distilled water. The λmaxor the absorption maximum of the drug was found to be 271nm. A linear response was observed in the range of 30- 150µg/ml with the regression coefficient of 0.999. The method was validated for different parameters as per the ICH (International Conference for Harmonization) guidelines. This method can be used for the determination of Tramadol Hydrochloride in quality control of formulation without interference of the excipients.[60]

8. Yogesh M *et al*., (2011) formulated transdermal drug delivery system of budesodine using Eudragit RLPO, RSPO, Ethyl cellulose, Poly Vinyl Pyrrolidine as polymers by mercury substrate method where DMSO was used as the penetration enhancers. Various *invitro* testing like drug permeation, moisture absorption was carried out on these patches[68].

9.Subash S *et al*., (2011) developed transdermal patches of Diltiazem Hydrochloride Hydroxy Ethyl Cellulose, Ethyl Cellulose and Eudragit RLPO in different proportions. Six batches of transdermal patches were prepared by solvent casting technique. The prepared patches were evaluated for physical character like share spherical, uniform in the shape and white in color. The obtained films evaluated for physico-chemical characteristics, In vitro and in vivo evaluation in albino mice. Higuchis plot studies revealed that the predominant mechanism of drug release was diffusion[69].

10.Yasuko Obata *et al;* (2010) developed ondansetron hydrogel system in transdermal patch using HEC and HPC as a bases, l-menthol as a penetration

enhancer and isopropanol (IPA), N-methyl-2-pyrrolidone and water as a solvent. These *invitro* and respionse linear studies indicated the feasibility of delivering Ondansetron transdermally[58].

11. Iman IS *et al.,* (2010) developed transdermal patches of chlorpheniramine maleate (CPM). They used rabbit ear skin membrane for *ex-vitro* diffusion studies and they used polymers like ethyl cellulose, cellulose acetate and poly vinyl pyrrolidone with different plasticizers such as propylene glycol and polyethylene glycol 400. They compare the CPM transdermal patch with CPM oral tablets and results showed that CPM transdermal patch has higher bioavailability than oral tablet of same dose, with lower plasma fluctuation and less administration frequency[63].

12. Samip.S *et al.,* (2010) developed transdermal patches of Papaverine hydrochloride prepared by the solvent casting method using EC: PVP, PVA: PVP & ERL-100: ERS-100 using different ratios. The formulation containing PVA: PVP shows faster release rate (hydrophilic polymers) compared to ERL-100: ERS-100 (hydrophobic polymers) or combination of hydrophilic &hydrophobic polymers (EC & PVP)[64].

13. Sivakumar et *al.,* (2010) developed patches of ketotifen fumarate with different different ratios of HPMC-E5 (hydrophilic) and EC (hydrophobic) as polymers, 5 % v/w of DMSO as penetration enhancer by solvent evaporation technique. The diffusion studies show maximum release is observed with HPMC-E5 alone [65].

14. Ammar H O *et al.,* (2009) developed polymeric matrix system for prolonged delivery of Tramadol hydrochloride. The current study is focused on the innovation of a novel, simple, monolayer, easy-to-use, cost-effective and aesthetically acceptable bioadhesive transdermal delivery system overcoming the defects of the conventional "patch" as carrier system for tramadol, ensuring its adequate delivery, along with the physicochemical evaluation of the designed formulations.
Monolithic tramadol matrix films of chitosan, different types of Eudragit® and binary mixtures of both were prepared. As a single- polymer film, chitosan film showed best properties except for somewhat high moisture uptake capacity, insufficient strength and rapid release and Permeation. Polymer blends were

monitored in order to optimize both properties and performance. Promising results were obtained, with chitosan–Eudragit® NE30D (1:1) film showing the most desirable combined, sufficiently rapid as well as prolonged release and permeation profiles along with satisfactory organoleptic and physicochemical properties [61].

15.Muhammad Naeem Aamir *et al*., (2009) developed the Tramadol microparticles by Non-solvent Addition Method and their *in vitro* Characterization. These microparticles with controlled-release characteristics were prepared by nonsolvent addition technique. Ethyl cellulose (EC), a hydrophobic polymer was employed to control and extend the drug release process. Dichloromethane (DCM) was employed as solvent for polymer and paraffin oil as non-solvent which induced phase separation. Microparticles of different polymer concentrations M1 (1:1), M2 (1:2) and M3 (1:3) were prepared. Among all these formulations, M3 presented superior and desirable characteristics, i.e., 79% entrapment efficiency, good micromeritic properties, smooth morphology, and more sustained effect on cumulative release. Zero order, first order, Higuchi, Hixson-Crowell, and Korsmeyer-Peppas kinetic models were applied to assess the mechanism and pattern of drug release from microparticles. Release of TmH was best fitted to Higuchi model because it presented highest values of correlation coefficient (R2 = 0.981) followed by zero-order kinetic model (R2 = 0.899)[62].

16.Nguyen Thien Hai *et al.,* (2008) developed drug-in adhesive transdermal patches of Benztropine (BZ). The patch was prepared by pressure sensitive adhesive. Diffusion tests were performed using rat skin. They showed good invivo activity suggesting the usage of patches in parkinsonism disease[56].

17. Sadhana PG *et al.,* (2005) developed Metaprolol Tartarate patches using eudragit RL and Hydroxy Propyl Methyl Cellulose. The developed patches were investigated for various *invitro* and *invivo* characteristics The diffusion profile was compared with that of standard[66].

18.Sanap GS *et al.,* (2008) developed transdermal drug delivery systems of indapamide by using solvent casting method. The study incorporated various

polymers like HPMC and EC glycerine, dibutyyl phthalated. various drug polymer ratios All the patrches were tested for various *invitro* tests and stability tests of the prepared patches[67].

19.Mi-Kyeong Kim *et al;* (2001) developed a reservoir-type transdermal delivery system of testosterone (TS) using an ethanol/water (70:30) cosolvent system as the vehicle. The maximum permeation rate achieved by 70% (v/v) of ethanol was further increased from 2.69 to 47.83 μg/cm^2 /h with the addition of 1.0% dodecyl amine as the skin permeation enhancer. This experimental patch showed comparable plasma concentration profiles in the *in vivo* study when compared with a commercial product[57]

Drug Profile:

Diclofenac[56]

Description: Diclofenac is a phenylacetic acid derivative and Non-steroidal Anti-Inflammatory drug (NSAID). NSAIDs inhibit cyclooxygenase COX-1 and COX-2 which are the enzyme responsible for producing Prostaglandins (PGs). PGs contribute to inflammation and pain signalling. Diclofenac, like other NSAIDs, is often used as first line therapy for acute and chronic pain and inflammation from a variety of causes.

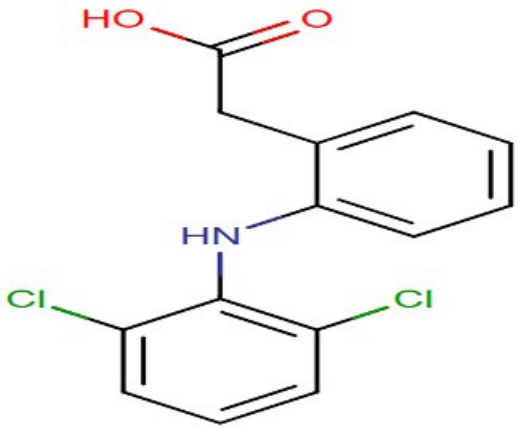


Figure 3.1 : Structure of Diclofenac

Synonyms

- Diclofenac
- Diclofenac acid
- Diclofenaco
- Diclofenacum

Weight

Average: 296.149

Monoisotopic: 295.016684015

Chemical Formula: C14H11Cl2NO2

Indication:

Diclofenac is indicated for use in the treatment of pain and inflammation from varying sources including inflammatory conditions such as osteoarthritis, rheumatoid arthritis, and ankylosing spondylitis, as well as injury-related inflammation due to surgery and physical trauma. It is often used in combination with misoprostol as a gastro-protective agent in patients with high risk of developing NSAID-induced ulcers. **Pharmacodynamics:** Diclofenac reduces inflammation and by extension reduces nociceptive pain and combats fever. It also increases the risk of developing a gastrointestinal ulcer by inhibiting the production of protective mucus in the stomach.

Mechanism of action:

Diclofenac inhibits cyclooxygenase-1 and cyclooxygenase -2, the enzymes responsible for production of prostaglandin (PG) G2 which is the precursor to other PGs. These molecules have broad activity in pain and inflammation and the inhibition of their production is the common mechanism linking each effect of diclofenac. PGE2 is the primary PG involved in modulation of nociception. It mediates peripheral sensitization through a variety of effects. PGE2 activates the Gq-coupled EP1 receptor leading to increased activity of the inositol trisphosphate/phospholipase C pathway. Activation of this pathway releases intracellular stores of calcium which directly reduces action potential threshold and activates protein kinase C (PKC) which contributes to several indirect mechanisms. PGE2 also activates the EP4 receptor coupled to Gs which activates the Adenylyl Cyclase/Protein Kinase A (AC/PKA) signaling pathway. PKA and PKC both contribute to the potentiation of transient receptor potential cation channel subfamily V member 1 (TRPV1) potentiation, which increases sensitivity to heat stimuli. It also activate tetrodotoxin-resistant sodium channels and inhibit inward potassium currents. PKA further contributes to the activation of the P2X3 purine receptor and sensitization of T-type calcium channels. The activation and sensitization of depolarizing ion channels and inhibition of inward potassium currents serve to reduce the intensity of

stimulus necessary to generate action potentials in nociceptive sensory afferents. PGE2 act via EP3 to increase sensitivity to bradykinin and via EP2 to further increase heat sensitivity. Central sensitization occurs in the dorsal horn of the spinal cord and is mediated by the EP2 receptor which couples to Gs. Pre-synaptically, this receptor increases the release of pronociceptive neurotransmitters glutamate, CGRP and substance P. Post-synaptically it increases the activity of AMPA and NMDA receptors and produces inhibition of inhibitory glycinergic neurons. Together these lead to a reduced threshold of activating, allowing low intensity stimuli to generate pain signals. PGI2 is known to play a role via its Gs-coupled IP receptor although the magnitude of its contribution varies. It has been proposed to be of greater importance in painful inflammatory conditions such as arthritis. By limiting sensitization, both peripheral and central, via these pathways NSAIDs can effectively reduce inflammatory pain. PGI2 and PGE2 contribute to acute inflammation via their IP and EP2 receptors. Similarly to β adrenergic receptors these are Gs-coupled and mediate vasodilation through the AC/PKA pathway. PGE2 also contributes by increasing leukocyte adhesion to the endothelium and attracts the cells to the site of injury. PGD2 plays a role in the activation of endothelial cell release of cytokines through its DP1 receptor. PGI2 and PGE2 modulate T-helper cell activation and differentiation through IP, EP2, and EP4 receptors which is believed to be an important activity in the pathology of arthritic conditions. By limiting the production of these PGs at the site of injury, NSAIDs can reduce inflammation. PGE2 can cross the blood-brain barrier and act on excitatory Gq EP3 receptors on thermoregulatory neurons in the hypothalamus.This activation triggers an increase in heat-generation and a reduction in heat-loss to produce a fever. NSAIDs prevent the generation of PGE2 thereby reducing the activity of these neurons.

Absorption: Diclofenac is completely absorbed from the GI tract but likely undergoes significant first pass metabolism with only 60% of the drug reaching systemic circulation unchanged Label,7,8. Many topical formulations are absorbed percutaneous and produce clinically significant plasma concentrations. Absorption is dose proportional over the range of 25-150 mg.7 Tmax varies between formulations with the oral solution reaching peak plasma concentrations in 1040min, the enteric coated tablet in 1.5-2h, and the sustained- and extended-release formulations prolonging Tmax even further. Administration with food has no significant effects on AUC but does delay Tmax to 2.5-12h.

Volume of distribution: Diclofenac has a total volume of distribution of 5-10 L or 0.1-0.2 L/kg. The volume of the central compartment is 0.04 L/kg. Diclofenac distributes to the synovial fluid reaching peak concentration 2-4h after administration. There is limited crossing of the blood brain barrier and cerebrospinal fluid concentrations only reach 8.22% of plasma concentrations. Doses of 50 mg delivered via intramuscular injection produced no detectable diclofenac concentrations in breast milk; however, metabolite concentrations were not investigated. Diclofenac has been shown to cross the placenta in mice and rats but human data is unavailable.

Protein binding: Diclofenac is over 99.7% bound to serum proteins, primarily albumin. It undergoes limited binding to lipoproteins as well with 1.1% bound to HDL, 0.3% to LDL, and 0.15% to VLDL.

Metabolism: Diclofenac undergoes oxidative metabolism to hydroxy metabolites as well as conjugation to glucuronic acid, sulfate, and taurine. Label,7 The primary metabolite is 4'-hydroxy diclofenac which is generated by CYP2C9. This metabolite is very weakly active with one thirtieth the activity of diclofenac. Other metabolites include 3'-hydroxy diclofenac, 3'-hydroxy-4'methoxy diclofenac, 4',5dihydroxy diclofenac, an acylglucuronide conjugate, and other conjugate metabolites.

Half-life: The terminal half-life of diclofenac is approximately 2 h,

Route of elimination: Diclofenac is mainly eliminated via metabolism.7,8 Of the total dose, 60-70% is eliminated in the urine and 30% is eliminated in the feces. No significant enterohepatic recycling occurs[59].

1.**EXCIPIENT PROFILE**

EUDRAGIT L 100

Nonproprietary Names:

NF: Methacrylic acid copolymer, polymeric methacrylates.

Functional Category: Film former, tablet binder.

Synonyms: Methacrylic acid; Polymethacrylate

Molecular Weight:

Molecular weight range: ≥ 100,000

Commercial Availability:

USA - Rohm Tech Inc.

UK - Dumas Chemicals (Tunbridge Wells) Ltd.

Method of Manufacture:

Polymerization of acrylic and methacrylic acids or their esters; e.g., butyl ester or dimethylaminoethyl ester.

Description:

Eudragit RL100 (10% trimethylammonium methacrylate chloride) is a copolymer of acrylic and methacrylic acid esters containing some quaternary ammonium groups, available as 12.5% ready-to-use solution in isopropanol and acetone

(60:40). Not more than a slightly yellow in color, usually clear or slightly turbid, it has the characteristic odour of solvents. Solvent-free granules contain ≥98% of the dried weight.

Typical Properties:

Density: 12.5: 0.825 g/cm

Solubility: Soluble in isopropanol and ethanol in combination with acetone or methylene chloride; also in methanol, chloroform, trichlorethylene, ethyl acetate and glycol monomethyl ether. Barely soluble in pure isopropanol or carbon tetrachloride. Insoluble in petroleum ether or light petroleum. Swells in aqueous media: the amount of water absorbed depends on the pH of the solution.

Viscosity: 5 to 15 cps (12.5% solution in isopropanol and acetone (60:40)).

Refractive index: 12.5: 1.383

Alkali value: 27

Stability and Storage Conditions:

Dry powder forms appear to be stable at room temperature. Dispersions are stable for about one year after manufacturing if stored at room temperature in tight containers. The dispersions are sensitive to extreme temperatures.

Safety: Acute toxicity studies have been performed in rats, rabbits and dogs. No toxic effects were observed at doses of dry lacquer substance ranging from 6-28 g/kg of body weight over a two-week period. Chronic toxicity studies were performed in rats over a period of 3 months. No significant changes were found in the animal organs.

Handling Precautions: The usual precautions must be employed when volatile solvents are used.

Applications in Pharmaceutical Formulation or Technology:

- ✓ Eudragit RL forms water-insoluble film coats for delayed-release products.
- ✓ Eudragit RL film coats are more permeable than those of Eudragit RS32[60].

2. SODIUM ALGINATE: Sodium alginate is a natural hydrophilic polysaccharide derived from seaweed. It is the sodium salt of alginic acid, a high molecular weight linear polymer consisting of D-mannuronic acid and L-glucuronic acid residues that are arranged in the polymer chain in blocks.

Empirical formula: $(C6H7O6Na)_n$

Description: Sodium alginate occurs as a white or buff colour coarse or fine powder which is odourless and tasteless. Aqueous solutions that form gel on the addition of small amount of soluble calcium salt.

Solubility: - Sodium alginate is slowly soluble in water, forming a viscous colloidal solution. It is insoluble in alcohol and in hydroalcoholic solutions in which the alcohol content is greater than 30% by weight. It is also insoluble in other organic solvents and in acids where the pH of the resulting falls below 3.

Empirical formula: $(C6H7O6Na)_n$

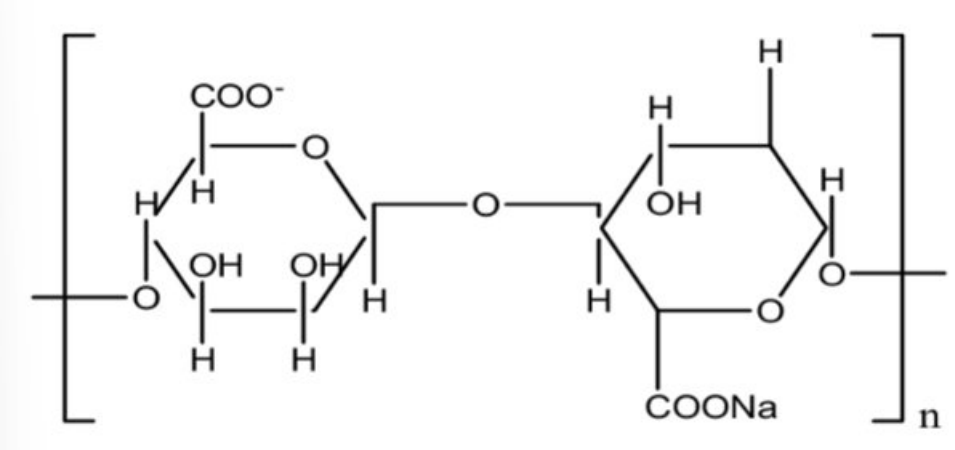


Figure 3.2 : Structure of Sodium Alginate

Viscosity: Various grades of sodium alginate are available yielding aqueous solutions of varying viscosity within a range of 20-400 centripoise (0.02 – 0.4 Pa.S) in 1% solution at 20C. Viscosity decreases of sodium alginate solutions above pH of 10.

Stability: Sodium alginate is hygroscopic, therefore the moisture content at equilibrium is a function of relative humidity. Dry storage stability is excellent when the powder is stored in well closed container at temperature of 25C. Incompatibility: Depending on the concentration, sodium alginate is incompatible with acridine derivatives, crystal violet phenyl mercuric nitrate, acetate, alcohol parabens in concentrations greater than 5% and heavy metals.

Immunogenicity and Biocompatibility: Biocompatibility and immunogenicity of material are important factors for successful application in carriers for drug delivery. Alginate is known to exhibit minimum cytotoxic effects and reduced haemolysis when contact with blood

Applications: Alginic acid and alginates such as propylene glycol algiante and sodium algiante are used as suspending and thickening agents in water miscible pastes, creams, gels, lotions. They are also used as stabilizers for oil in water emulsions and as binding and disintegrating agents in tablets. Several properties of alginate, such as its biodegradability, non-toxicity, biocompatibility and ability to form gel with a variety of cross-linking agents in mild and aqueous conditions, make it a useful carrier for controlled delivery of a variety of drugs and other biologically active agents.

3.CHITOSAN

Chitosan a natural polysaccharide, is being widely used as a pharmaceutical excipient. It is obtained by the partial deacetylation of chitin. Chitosan [α-(1Æ4)2-amino-2-deoxy-β-d-glucon] is synthesized by deacetylation of chitin. The basic process of the manufacture of chitosan involves the removal of proteins and minerals such as calcium carbonate and calcium phosphate by treating with alkali and acid respectively. The biomedical grade/purified chitosan is prepared by repeating deacetylation process. Pharmaceutical grade chitosan is deacetylated between 90% and 95% and the food grade between 75% and 80%.

Empirical Formula: (C6H11O4N)n

Physicochemical Properties: Chitosan is a linear polyamine with number of amino groups that are readily available for chemical reaction and salt formation with acids. Important characteristics of chitosan are its molecular weight, viscosity, degree of deacetylation, crystallinity index number of monomeric unit (n) water retention value

and energy of hydration. Chitosan has a high charge density, adheres to negatively charged surfaces and chelates metal ions. Due to high molecular weight and a linear unbranched structure, chitosan is an excellent viscosity in an acidic environment. It behaves as pseudoplastic material exhibiting a decrease in viscosity with increasing rates of shear. The viscosity of a solution increases with an increase in chitosan concentration, decrease in temperature and with increasing degree of deacetylation. It is biodegradable and non-toxic and has reactive hydroxyl group and amino groups that can be modified chemically for various applications for drug delivery.

Solubility:

Chitosan is insoluble in water and is soluble in variety of dilute acid solutions, which can convert glycosamine unit to a soluble R-NH3+ and interacts with polyanionic counter ions. The hydrophilicity of the polymer, due to amine functionalities is most repeat units, makes polymer soluble in dilute acid solutions and often yields a rubbery hydrogel in water.

Gel Properties:

Chitosan exhibits excellent compatibility with organic compounds as cationic dyes and surfactants, starches, quarternary ammonium salts and with most cationic and non-ionic polymers, multivalent anions easily cross-linked with chitosan to form gels and precipitates. The cationic nature permits it to form complexes with oppositely charged drugs/ excipients, thereby altering the physicochemical characteristics of the formulation. This ionotropic gelling property of chitosan with various counter ions is given below.

Applications:

The unique combination of biological, physical and chemical properties, chitosan is widely used in various fields like biomedical, pharmaceutical food and agriculture, cosmetic and water treatment.

Pharmaceutical Applications:

✓ Chitosan is used in conventional formulations like compressible tablets, controlled release matrix tablets, wet-granulation, gels, films, emulsions, wetting and coating agent.

✓ Chitosan also used for novel applications in microspheres and microparticulate drug delivery, bioadhesion, vaccine delivery, peptide delivery, transdermal and transmucosal drug transport, gene delivery, nanotechnology.

4. PROPYLENE GLYCOL

Description:

Propylene glycol is a synthetic liquid substance that absorbs water. Propylene glycol is also used to make polyester compounds, and as a base for deicing Department of Pharmaceutics, MVM College of Pharmacy Methodology solutions. Propylene glycol is used by the chemical, food, and pharmaceutical industries as an antifreeze when leakage might lead to contact with food. The Food and Drug Administration (FDA) has classified propylene glycol as an additive that is "generally recognized as safe" for use in food. It is used to absorb extra water and maintain moisture in certain medicines, cosmetics, or food products. It is a solvent for food colors and flavors, and in the paint and plastics industries. Propylene glycol is also used to create artificial smoke or fog used in fire-fighting training and in theatrical productions. Other names for propylene glycol are 1,2dihydroxypropane, 1,2-propanediol, methyl glycol, and trimethyl glycol. Propylene glycol is clear, colorless, slightly syrupy liquid at room temperature. It may exist in air in the vapor form, although propylene glycol must be heated or briskly shaken to produce a vapor. Propylene glycol is practically odorless and tasteless[61].

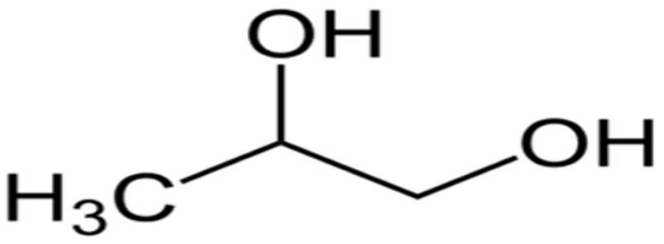


Figure 3.3 : Structure of Propylene glycol

Synonyms

propylene glycol

1,2-propanediol

propane-1,2-diol

1,2-Propylene glycol

Molecular Formula

C3H8O2

Molecular Weight

76.09 g/mol